A Detector's Corroboration Needs a Control Arm: A Case Study in Coastal Upwelling

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Abstract. A detector that corroborates one signal against a second, independent observation is usually reported by its agreement rate alone, which is easy to mistake for evidence of causation. We report a case inside a real, shipping coastal-upwelling detector where adding a control arm changes that reading. The detector implements Bakun's wind-driven Ekman-transport index, with a coastal normal derived from the ocean mask gradient rather than a coastline dataset, and corroborates it against an independent cold sea-surface-temperature (SST) anomaly scored against a percentile climatology. On the live global field, measured 2026-08-17, 19.9% of upwelling-favourable coastal cells carried a cold anomaly for the season. Scored identically on a control arm — coasts where the wind is downwelling-favourable, where a cold anomaly is not expected — the same test returns 17.2%, a gap of only 2.7 percentage points that narrows to 0.2 at a stronger, percentile-based tier. A follow-up experiment this platform's own prior account named as the natural next step — computing the index against a real, backfilled 1.2-day trailing mean of wind forcing rather than one instantaneous snapshot, to test whether integrating wind over time strengthens the signal — has since been built and measured, and does not widen the gap; its movement is not clearly distinguishable from the day-to-day spread the identical, unmodified configuration already shows across separate measurement dates. The detector's hemisphere sign convention, independently verified here by running the platform's test suite directly, is protected by 39 passing regression tests, including an adversarial fixture built to catch the one error that would report every upwelling coast on Earth as downwelling. That guarantee is orthogonal to corroboration strength, and the platform now returns a control fraction with every response rather than leaving a reader to compute one.

Keywords: coastal upwelling, Bakun index, control arm, detector corroboration, sea surface temperature, Ekman transport

1 Introduction

A detector that corroborates one signal against a second, independent observation almost always reports the same number: the fraction of its own detections the second signal agrees with. That fraction is easy to compute, easy to state,

and, read alone, easy to mistake for evidence that the first signal caused the second. It is not evidence of that on its own, because a reader has no way to tell a genuine mechanism from a coincidence without knowing how often the same corroborating signal would appear anyway, in a place the first signal is not supposed to hold. This paper reports a case inside a real, currently shipping detector where making that comparison changes the finding from a confident-sounding confirmation to a barely detectable one, and where a natural follow-up experiment — built and measured after the fact — does not rescue it.

The detector is MarisAI's coastal upwelling index, an implementation of Bakun's wind-driven Ekman-transport formulation [2]: wind stress at the sea surface drives a transport 90 degrees to the right of the wind in the northern hemisphere and to the left in the southern, and where that transport points offshore, the water it removes is replaced from below. The index alone says only that the wind is favourable for upwelling, not that cold, nutrient-rich water has actually surfaced — stratification, topography, and relaxation all intervene between a favourable wind and an observed response. The platform corroborates the index against an independent signal that can, in principle, close that gap: a coincident cold sea-surface-temperature (SST) anomaly, scored against a per-cell, per-day-of-year percentile climatology fitted on the NOAA OISST record. On the live global field, measured 2026-08-17, 19.9% of upwelling-favourable coastal cells carried a cold anomaly for the season. Read alone, that number is a confirmation. Scored identically on downwelling-favourable coasts — water the same test is not expected to find cold at all — the control arm returns 17.2%, a gap of only 2.7 percentage points, and the stronger, percentile-based tier of the same test narrows the gap further still, to 0.2 points.

The system's own prior account of this result, recorded in the detector's source code and in this platform's own research notes, names an unfinished experiment as the natural next step: both arms of the control test score one instantaneous wind snapshot against one instantaneous SST reading, while upwelling is physically understood to respond to wind forcing integrated over days, not hours. This paper verifies, against the code and the test suite directly rather than against those notes, that this experiment has since been built and run — and reports what it actually shows: integrating the wind over a real, back-filled 1.2-day trailing window does not widen the gap between the favourable and control arms. Read beside the natural day-to-day variation the same, untouched configuration already shows across three separate live measurements, the movement from averaging the wind is not distinguishable from that background variation either. The honest finding is not "the wind index is meaningless" — the underlying physics is well established and the index's own hemisphere sign convention is pinned by a regression test built specifically to catch the one error that would make every coast look like its opposite — but that a single day's corroboration number, with or without a longer wind window, is not by itself the evidence of causation it is easy to read it as.

1.1 Contributions

1. A **verified account of a shipping detector's control-arm design**: the coastal upwelling index's construction (Bakun's Ekman-transport formulation), a coastal-normal derivation built from the ocean mask rather than from a coastline dataset (because none is available to this platform), and an SST corroboration test scored at two tiers against a percentile climatology, with a control fraction computed and returned as a shipped field rather than left for a caller to compute.
2. A **re-measurement of the headline corroboration numbers** and an independent confirmation, by running the platform's own test suite directly, that the hemisphere/eastern-boundary sign convention the index depends on is covered by 39 currently passing regression tests (§4).
3. A **second, follow-up experiment** — verified to exist and to have been measured, contrary to an assumption a reader of this platform's own research notes would otherwise carry into this paper — that tests whether integrating the wind input over a real, multi-day trailing window widens the control-arm gap. It does not, and §4 reports the movement beside the natural day-to-day spread of the same, unchanged configuration, rather than in isolation.
4. An explicit **evidence grading** of every claim, following the Artifact/Dated/Artifact-plus-dated convention this research program already uses [5]: the corroboration percentages and the wind-history follow-up are each single, dated, live measurements of moving fields, and are reported as such rather than as standing facts, while the control fraction's existence as a shipped field and the sign-convention regression suite are each independently verifiable right now.

2 Related Work

Bakun's index and its known limitations. Bakun's coastal upwelling index, formalised in the wind-driven-intensification literature this platform's implementation follows [2], has long depended on a single, often simplified assumption about which way is offshore: the classical index is typically computed against one fixed coastline angle per region. Jacox et al. [4] revisit exactly this assumption for the U.S. West Coast, building an improved index from a high-resolution coastline dataset that varies the coastal angle along the shore rather than fixing it, and show the fixed-angle assumption measurably changes the computed transport in stretches of complex coastline. This platform reaches the same goal — a per-cell offshore direction rather than one fixed bearing — by a different route, described in §3: no coastline geometry service is available to it, so the coastal normal is derived from the gradient of the ocean mask the currents field already carries, rather than from a coastline database. The two approaches share a diagnosis (a single fixed coastline angle is not good enough) and differ in what each has available to fix it with.

Upwelling systems and why the corroborating signal matters. Chavez and Messié's comparative review of the world's major eastern-boundary upwelling ecosystems [3] documents the physical chain this paper's corroboration test is built to shortcut: favourable wind stress drives Ekman transport, which drives upward advection of cold, nutrient-rich water, which in turn drives the biological productivity these ecosystems are known for. Every step in that chain is a place the chain can be interrupted — by stratification, by relaxation, by the wind not yet having blown long enough — which is exactly why a wind-derived index and an SST-based corroboration are two different claims and not, as §3 describes, one claim measured twice.

Control and placebo arms outside this platform's own domain. The general discipline of testing a hypothesised effect against an arm where it is not expected to hold is well established outside atmospheric and ocean science. Abadie, Diamond, and Hainmueller's synthetic control method [1] builds a placebo test directly into a comparative case-study design: an estimated treatment effect is checked against the same estimator applied to units that were never treated, and an effect indistinguishable from what the placebo units show on their own is not evidence of a real intervention effect. The construction here is the same shape, applied to a geophysical detector rather than a policy intervention — downwelling-favourable coasts play the role of units where the treatment (upwelling-favourable wind) did not occur, and the corroborating signal (a cold anomaly) is scored identically on both arms.

Overinterpreting agreement in atmospheric science specifically. Wilks [7] documents, within the discipline this paper's detector belongs to, how a raw count or rate computed over many spatial locations is routinely reported and interpreted more confidently than its own base rate supports, and argues for reporting a comparison against chance rather than the raw statistic alone. This paper's contribution is a single, concrete instance of exactly that argument, applied to a wind/SST corroboration test rather than to a gridded significance map, with the base-rate comparison computed from a downwelling control arm rather than from a randomisation test.

This research program's own prior treatment. This case was first catalogued as one of eleven silent, plausible, signed-direction failure instances found across this same platform [5], where it is instance I.11, summarised in a few sentences with the same 2026-08-17 headline numbers reported in §4. This paper is the fuller methodological treatment of that single instance — the index construction, the coastal-normal derivation, the corroboration test's design, and, unlike that catalogue entry, the wind-history follow-up experiment that had not yet been run when it was written. A separate, contemporaneous companion draft in this research program (not yet assigned a title or conference wrappers) analyses a different weakness in the same corroboration test — that the SST observation and the climatology it is scored against can come from different products, which measurably inverts the same test's strong tier — and, for a different purpose,

reports the identical 2026-08-28 wind-history measurement analysed here: that draft uses it to rule out instantaneous wind sampling as an alternative explanation for its own cross-product finding, while this paper uses the same run to ask whether temporal wind averaging widens the control-arm gap this paper's own §4 establishes. §5 states this overlap explicitly rather than leaving two papers to report the same four numbers without saying they are one measurement.

3 Methods

Bakun's index, as implemented. Wind stress is computed from the bulk formula $\vec{\tau} = \rho_{\text{air}} C_d |\vec{U}| \vec{U}$, with air density $\rho_{\text{air}} = 1.225\,\text{kg/m}^3$ and a drag coefficient $C_d = 1.3 \times 10^{-3}$, the Large and Pond mid-range constant [6] rather than a wind-speed- or stability-dependent value — the standard simplification for an index rather than a flux budget. Depth-integrated Ekman transport is then $\vec{M} = (\tau_y/(\rho_w f), -\tau_x/(\rho_w f))$, with seawater density $\rho_w = 1025\,\text{kg/m}^3$ and the Coriolis parameter $f = 2\Omega \sin(\text{latitude})$, $\Omega = 7.2921 \times 10^{-5}\,\text{rad/s}$. The hemisphere asymmetry — transport 90 degrees to the right of the stress in the north, to the left in the south — falls directly out of the sign of f in this formula rather than from an explicit latitude branch, which matters for §4's regression test: a hard-coded rotation would pass a northern-hemisphere test and fail only in the south, an easy mistake to ship unnoticed. Because $f \to 0$ at the equator, the transport formula diverges there and equatorial upwelling is in any case a different, divergence-driven mechanism rather than a coastal one; cells within 5 degrees of the equator are excluded outright. The index itself is the transport's projection onto the local offshore normal, index $= \vec{M} \cdot \hat{n}_{\text{offshore}}$: positive is upwelling-favourable, negative is downwelling-favourable, and getting this sign backwards would report every upwelling coast on Earth as downwelling while still producing an entirely plausible-looking field.

Deriving a coastal normal without a coastline dataset. Projecting a transport vector onto "offshore" requires knowing which way offshore is, and no service on this platform supplies coastline geometry. The normal is instead derived from the ocean mask the surface-currents field already carries (the Copernicus physics product is NaN over land): the mask is smoothed over 3 cells, and the gradient of the smoothed mask points from land into water by construction, since "how much ocean is nearby" increases moving offshore. A raw, unsmoothed 0/1 mask has a gradient only in the single cell touching land, which is why the smoothing step exists at all — without it, every other cell in the coastal band would have no defined normal. The wind field is resampled onto this mask's grid rather than the reverse, because the wind product's own coverage extends over land as well as ocean, so its coverage edge is not a coastline at all. Confidence in the resulting normal is the magnitude of the smoothed gradient, scaled against the steepest gradient the grid produces anywhere; a cell whose confidence falls below 0.05 — an enclosed sea, a cell wedged between two coastlines, a one-cell island — is dropped rather than assigned a bearing that would not mean anything. This is a genuinely coarse construction at the currents product's roughly

0.25-degree resolution, below which a headland, bay, or fjord is not resolved; `coastline_confidence` is reported with every response for exactly this reason, and only cells within 2 grid cells of land (≈55 km at this resolution) are scored at all, since the index is defined per metre of coastline and applying it mid-ocean would project onto a normal that does not exist.

SST corroboration against a percentile climatology. The corroborating signal is a per-cell, per-day-of-year climatology fitted on the OISST daily record over 1991–2020, pooled over a ±5-day window around each calendar day, giving a mean and the 10th/90th percentiles at every ocean cell and day of year. A fresh SST observation is scored against this baseline two ways: a weak, mean-based test (the observed anomaly is at least 0.5 degrees Celsius below the climatological mean) and a strong, percentile-based test (the observation falls below its own seasonal 10th percentile — a tail test defined by the local distribution rather than by a hand-chosen constant). The two tests are reported as separate corroboration states rather than collapsed into one, and the SST field's own timestamp and lag are published beside the result rather than folded into the wind timestamp, because OISST publishes daily with a lag that reached 16 days when measured on 2026-08-17 — well inside the 30-day limit past which corroboration is withheld outright as too stale to say anything about the current wind, but far from simultaneous with it. A coastal cell outside OISST's coverage is reported as `sst_unavailable`, a state distinct from "not corroborated": a coverage gap is not a finding about the ocean, and treating it as one would report every uncovered cell as a refutation it is not.

The control arm. Following this platform's own stated discipline that a corroboration rate is not evidence of causation without a control arm [5], the same cold-anomaly test that scores upwelling-favourable cells is scored identically on downwelling-favourable cells, where a coincident cold anomaly is not physically expected at all. A warm anomaly under downwelling-favourable wind is not treated as the symmetric confirmation it might look like — downwelling suppresses a cold signal rather than producing a warm one, and a warm surface has a dozen causes unrelated to the coast — so downwelling cells are scored only on whether they are anomalously *cold*, the one comparison that is a fair test of whether the favourable arm's own cold cells mean anything beyond coincidence. Only the weak tier's control fraction — named `control_cool_fraction` rather than a name implying it measures downwelling itself, since its job is to be a denominator of belief, not another finding about downwelling coasts — is a field the live API actually returns with every response; the strong-tier control comparison and the mean-anomaly comparison reported in §4 are ad hoc measurements from the platform's own diagnostic scripts, not values exposed by the shipped endpoint.

The wind-history follow-up. Both arms of the control test as described above score one instantaneous wind snapshot against one instantaneous SST reading, while upwelling is understood physically to respond to wind forcing integrated

over days rather than to a single hourly sample. A trailing, multi-day mean of the Ekman transport itself (not of the raw wind vector, since stress is quadratic and directional and averaging raw wind first can cancel real forcing a fluctuating-but-strong wind actually applied) is kept on a fixed 1-degree grid in an in-process ring buffer, retained for up to four days. Testing this required a real, multi-day backfill of historical wind timesteps, since a freshly started process has no history to average over; the backfill obtained 30 real hourly timesteps spanning 1.2 days, short of the originally targeted 3-day window, because the underlying wind cache carries a documented gap of 24 consecutive empty hourly placeholders that the backfill routine has to look back 30 hours to route around. A trailing mean refuses to be computed at all unless the collected samples span at least 50% of the requested window, so a short window is reported honestly as a short window rather than silently relabelled. The same coastal-normal, band-exclusion, and corroboration logic used for the instantaneous case is shared by construction between the two code paths, so a comparison between an instantaneous run and a windowed one is a comparison of the wind input alone, never of two slightly different definitions of "upwelling-favourable".

Evidence grading. Every measurement in §4 is graded following the Artifact/Dated convention this research program already uses [5]: *Artifact* for a source-code constant or a currently-passing regression test, reproducible on demand and not subject to drift; *Dated* for a single, live measurement of a moving field, recorded with its date and its reproduction script but with no output file committed; and *Artifact + Dated* where a tested, shipping mechanism is paired with a headline number that is itself a Dated measurement. The 2026-08-17 corroboration percentages and the 2026-08-28 wind-history numbers are each Dated; the control fraction's existence as a shipped field, the hemisphere/eastern-boundary regression suite, and the wind-history infrastructure itself are each Artifact.

4 Results

Table 1. The corroboration test at both tiers, favourable versus control, measured live 2026-08-17. Percentages; the gap is favourable minus control. [DATED]

Tier	Favourable	Control	Gap
Weak (cool anomaly, $\leq -0.5°C$)	19.9%	17.2%	2.7 pts
Strong (below seasonal p10)	4.1%	3.9%	0.2 pts

The headline gap is narrow, and narrows further at the stronger test. Table 1 reports the live global measurement. At the weak, mean-based tier, 19.9% of upwelling-favourable coastal cells carried a cold anomaly for the season, against

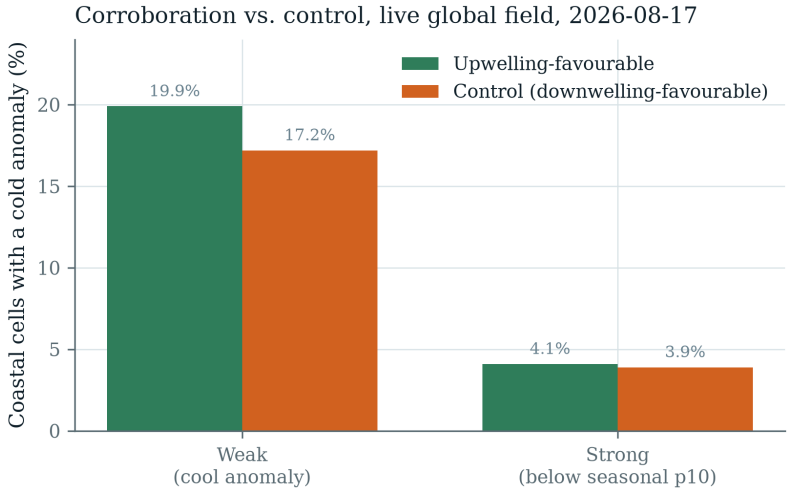


Fig. 1. The same two tiers as Table 1, plotted: favourable coasts (green) against the downwelling-favourable control arm (orange). Neither tier separates cleanly from its own control.

17.2% of downwelling-favourable cells — a gap of 2.7 percentage points. At the strong, percentile-based tier the gap narrows to 0.2 points (4.1% against 3.9%). The mean SST anomaly itself is *positive* on both arms that day — +0.91°C under favourable wind, +0.60°C under downwelling — so the coastal band as a whole read warm for the season regardless of wind regime, and the favourable arm was not even the cooler of the two. No formal significance test exists anywhere in this codebase for either gap; the honest statement is that both gaps are small relative to their own percentages, not that either is statistically indistinguishable from the other in a sense this measurement was designed to test.

The sign convention is independently regression-tested. Running the platform's own test suite directly (`uv run pytest tests/test_upwelling.py tests/test_wind_history.py -q`) on 2026-09-10 passes all 39 tests (32 in `test_upwelling.py`, 7 in `test_wind_history.py`). The test most directly relevant to this paper's own construction is a deliberately adversarial fixture: a coastline placed with land in the *high*-longitude columns and open ocean to the west, called an eastern boundary — the geometry of California, Peru, Namibia, and Somalia — checked against an equatorward wind that must upwell there, and mirrored in latitude to confirm the southern hemisphere reverses sign rather than repeating the northern result. Getting the coastline's orientation backwards in this fixture is the one error that would report every upwelling coast on Earth as downwelling while still producing a field that looks entirely plausible, exactly the failure mode §3 warns the sign convention is exposed to.

Table 2. The wind-history follow-up, read against the natural day-to-day spread of the same, unchanged matched configuration (OISST scored against its own OISST-fitted climatology, instantaneous wind unless noted). Contrasts are favourable-minus-control fractions, not percentages, and are not comparable to Table 1's units. [DATED throughout; the 1.2-day trailing mean row is Artifact-backed infrastructure with a Dated value.]

Date	Wind input	Weak contrast	Strong contrast
2026-08-17	instantaneous	0.026	0.002
2026-08-25	instantaneous	0.027	-0.001
2026-08-28	instantaneous	0.0234	-0.0061
2026-08-28	1.2-day trailing mean	0.0182	-0.0049

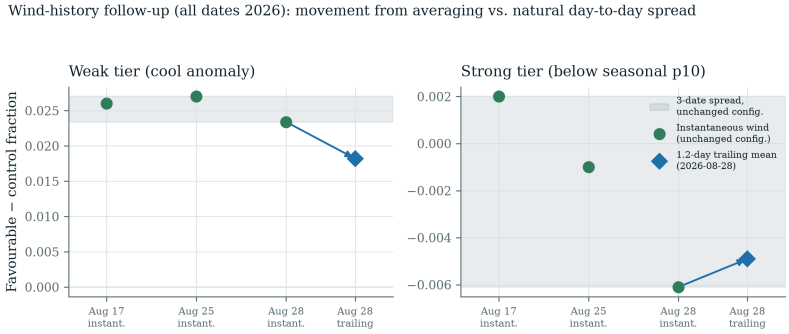


Fig. 2. Each panel plots one tier's contrast across four measurements: three dates of the identical, unchanged matched configuration (circles) and the 2026-08-28 trailing-mean arm (diamond), connected to its own day's instantaneous value (arrow). The shaded band spans the three-date range of the unchanged configuration – the natural day-to-day spread the wind-averaging movement has to be read against.

The wind-history follow-up does not widen the gap. Research notes associated with this detector, written before this paper's own verification pass, name a rolling wind history as the natural next experiment and describe it as not yet built. It has since been built (§3) and measured, on 2026-08-28, against a real backfilled 1.2-day window. Table 2 reports the result beside two earlier measurements of the identical matched configuration, with nothing about the method changed between them, so the table also shows how much this configuration moves on its own from one live snapshot to the next. On the weak tier, the trailing mean moves the contrast from 0.0234 to 0.0182, a movement of 0.0052 — toward zero, not away from it, and somewhat larger than the 0.0036 spread the same unchanged configuration already shows across the two earlier dates in the table. On the strong tier the trailing mean moves the contrast from -0.0061 to -0.0049, a movement of only 0.0012 — smaller than the 0.0081 spread the unchanged configuration already shows across its own three measurement dates, which itself *changes sign* (0.002, -0.001, -0.0061) with nothing about the method varied at all. Averaging the wind over the available window neither widens the gap this paper's control arm shows to be narrow, nor moves by more than the day-to-day noise the same test already exhibits when nothing is changed.

5 Discussion

A corroboration rate needs its control before it means anything. 19.9% read alone is a confident-sounding number: a fifth of every upwelling-favourable coast was cool for the season. Scored beside 17.2% on coasts where a cold anomaly is not expected, the same number shows an agreement barely wider than what the same test finds where it should find nothing. This is the paper's central point stated as plainly as the measurement supports: the wind index and the SST observation are two real, independently computed claims about the same coast, and their coincidence on 2026-08-17 is weak evidence connecting them, not because either measurement is wrong, but because a single day's snapshot at this resolution and this SST lag does not carry much evidential weight either way.

What the wind-history follow-up actually shows. It would overstate this paper's own evidence to read Table 2 as proof that no wind-averaging window would help; it is one measurement at one window length, 1.2 days against an originally targeted 3, obtained because that is what a real backfill of the platform's own wind cache produced. What it does show, honestly stated, is that this specific attempt did not widen the gap, and that the size of its effect is not clearly distinguishable from the day-to-day variation the identical, unmodified configuration already displays when measured on different dates with nothing about the method changed. The strong tier's own cross-date instability — positive on 2026-08-17, negative on 2026-08-25, more negative still on 2026-08-28, with no change to the method between any of those three measurements — is itself informative: it shows the strong tier is dominated by noise at the scale these measurements operate at, which is exactly the condition under which asking

that tier to demonstrate a wind-averaging effect is unlikely to succeed regardless of whether integrating the wind over days is physically the right idea. Table 2's honest reading is "tried once, real data, did not help" — not "proven not to help at any window length."

One run, read for two different questions. The 2026-08-28 measurement in Table 2's last two rows is the identical run a contemporaneous companion draft in this research program analyses as its own control measurement, for a different purpose: that draft holds the SST product and its climatology fixed throughout and uses this same instantaneous-versus-trailing-mean comparison to rule out instantaneous wind sampling as an alternative explanation for a cross-SST-product mismatch it measures separately. Both readings are compatible and draw on the same four numbers; neither supersedes the other, since the questions are different — whether wind averaging explains an unrelated cross-product inversion, versus whether wind averaging rescues this paper's own control-arm gap. Reporting both without saying they share a measurement would understate how much independent evidential weight one dated run of a live pipeline actually carries.

The regression test protects the sign, not the corroboration strength. 39 passing tests, including the adversarial eastern-boundary/hemisphere fixture described in §4, establish that the index's own sign convention is correct and will stay correct under refactoring. That is a different, narrower guarantee than "the corroboration is strong," and this paper is explicit that the two should not be conflated: a detector can have a perfectly reliable sign convention and still show a control-arm gap too narrow to support a causal claim on a given day, because the sign convention and the corroboration strength are answers to different questions. The platform's response to the second, weaker finding is not to remove the corroboration — a real, if weak, coincidence is still worth reporting — but to compute and return `control_cool_fraction` with every response, so a caller reading the corroborated fraction is never more than one field away from the number that says how much of it might be chance.

6 Limitations

Both of this paper's central measurements are single, dated, live observations of moving fields, not repeated-sampling or formally tested results: no significance test exists anywhere in this codebase for the 2026-08-17 headline gap, the three-date spread in Table 2, or the wind-history movement, and this paper reports raw gaps and spreads rather than a p-value or confidence interval it cannot honestly produce. The wind-history follow-up specifically reached only a 1.2-day window against the 3-day window originally targeted, limited by a real, documented gap in the underlying wind cache rather than by a design choice; a longer window, or repeated sampling across several separate days and seasons for either the headline measurement or the follow-up, remains untried. The infrastructure to run either — `services/wind_history.py`, `upwelling.detect_from_history`, and

the backfill script — is already built and tested, which makes either extension a re-run rather than a re-build, but neither has been done as of 2026-09-10. Only the weak tier's control fraction is a field the live API exposes; a caller relying on the shipped response alone sees the weak-tier comparison this paper reports but not the strong-tier or mean-anomaly evidence, both of which currently exist only as dated numbers in the detector's own source comments. The third, chlorophyll-based corroborating leg the platform's own documentation describes as a long-standing goal remains blocked on the same missing input it always was: there is no resident chlorophyll field and no long-record chlorophyll baseline to fit a climatology on, so a three-way corroboration (wind, SST, and biological response) is not yet possible to build, let alone measure. Finally, this is one detector, one SST product, one climatology, and one geographic scope (the global coastal band at ∼0.25-degree resolution); whether a control-arm gap this narrow is characteristic of wind-driven upwelling corroboration generally, or specific to this platform's product choices and measurement dates, is not established by a single running system.

7 Conclusion

A wind-derived upwelling index corroborated against an independent cold-SST observation reads, on its own, as confirmation: 19.9% of favourable coastal cells were anomalously cool for the season on the day measured. Scored identically on a control arm where that signal is not expected — downwelling-favourable coasts — the same test returns 17.2%, a gap of 2.7 points that narrows to 0.2 at the stronger, percentile-based tier, with the mean SST anomaly warmer than the seasonal average on both arms that day. A follow-up experiment this platform's own prior account named as the natural next step — integrating the wind input over a real, multi-day trailing window rather than scoring one instantaneous snapshot — has since been built and measured, and does not widen the gap; its movement is not clearly distinguishable from the day-to-day variation the identical, unmodified configuration already shows across separate measurement dates. Detector corroboration is usually reported as a single agreement rate and left there. The discipline this paper applies, twice, is cheap, computed from data already in hand, and available to any detector that corroborates one signal against a second: score the same test where the effect is not expected, and read the gap, not the level.

Reproducibility

The detector is `backend/services/upwelling.py`; the corroboration test and its control arm are `UpwellingField.corroboration()`, consumed by `GET /api/ocean/upwelling{,/point,/`
The 2026-08-17 headline measurement and the 2026-08-17/2026-08-25 matched-configuration values in Table 2 are recorded directly in that module's and in `backend/services/sst_anomaly.py`'s own module docstrings; both are reproducible on demand by running `scripts/measure_sst_corroboration.py` against

a live Copernicus/OISST snapshot, but no output file is committed for either date. The hemisphere/eastern-boundary regression suite is `backend/tests/test_upwelling.py` (32 tests) and `backend/tests/test_wind_history.py` (7 tests), all 39 passing as of 2026-09-10 via `uv run pytest tests/test_upwelling.py tests/test_wind_history.py -q`. The wind-history follow-up is `backend/services/wind_history.py` and `upwelling.detect_from_history` backfilled by `scripts/backfill_wind_history.py` and measured by `scripts/measure_wind_history_corro` its 2026-08-28 numbers are recorded in `sst_anomaly.py`'s docstring and in `DONE.md`'s "A rolling wind history" entry, with no stored output file committed. Every number in this paper was re-verified against those sources, and every relevant test was re-run, as of 2026-09-10.

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